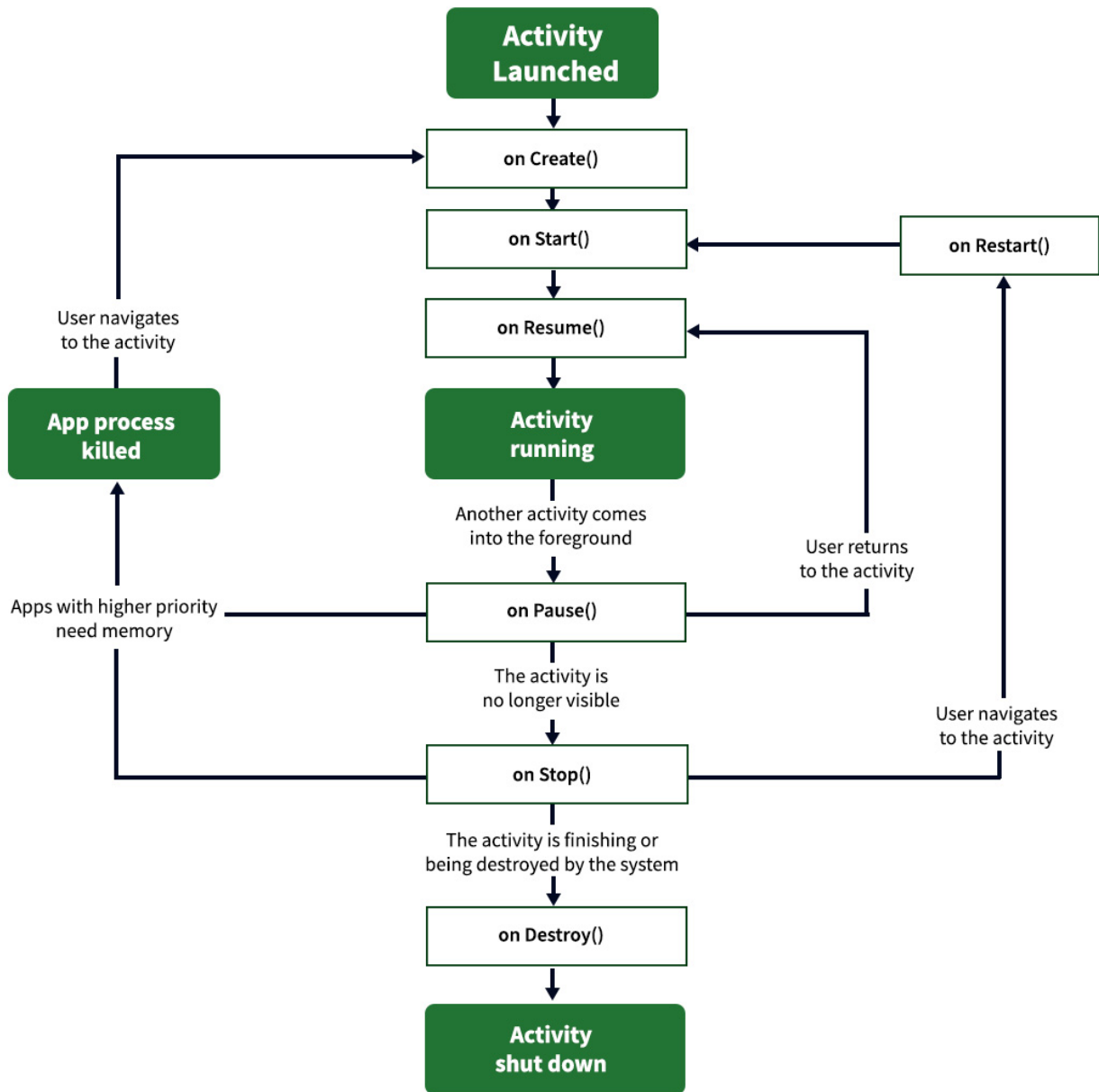


Class 6

Activity 生命週期



Activity Lifecycle in Android

```
override fun onStart() {  
    super.onStart()  
}
```

```
override fun onResume() {  
    super.onResume()  
}
```

```
override fun onPause() {  
    super.onPause()  
}
```

```
override fun onStop() {  
    super.onStop()  
}
```

```
override fun onDestroy() {  
    super.onDestroy()  
}
```

```
override fun onRestart() {  
    super.onRestart()  
}
```

使用 lateinit

```
class MainActivity : AppCompatActivity() {  
  
    private lateinit var nextButton: Button  
    private lateinit var nameTextView: TextView  
  
    override fun onCreate(savedInstanceState: Bundle?) {  
        agreeCheckBox = findViewById(R.id.agreeCheckBox)  
        nextButton = findViewById(R.id.nextButton)  
    }  
}
```

使用 onSaveInstanceState 保存數據

```
private var number = 0  
  
override fun onSaveInstanceState(outState: Bundle) {  
    super.onSaveInstanceState(outState)  
    outState.putInt("number", number)  
}
```

在 onCreate 恢復 onSaveInstanceState 保存的數據

```
override fun onCreate(savedInstanceState: Bundle?) {  
  
    if(savedInstanceState != null) {  
        number = savedInstanceState.getFloat("number", 0f)  
        textView.text = number.toString()  
    }  
}
```

events 事件

Checkbox 點擊狀態發生改變時

```
val agreeCheckBox: CheckBox = findViewById(R.id.agreeCheckBox)
agreeCheckBox.setOnCheckedChangeListener { _, isChecked ->
    if(isChecked) {
        nextButton.isEnabled = true
    } else {
        nextButton.isEnabled = false
    }
}
```

EditText 輸入改變時

```
nameEditText.doOnTextChanged { text, _, _, _ ->
    nameTextView.text = text
}
```

RadioButton 選擇改變時

```
radioGroup = findViewById(R.id.radioGroup)
radioGroup.setOnCheckedChangeListener { _, checkedId ->
    when(checkedId) {
        R.id.radioButton1 -> {
            nameTextView.text = "Radio Button1 選中了"
        }
        R.id.radioButton2 -> {
            nameTextView.text = "Radio Button2 選中了"
        }
        else -> {
            nameTextView.text = "Radio Button3 選中了"
        }
    }
}
```

使用 Shape 圖形



為 TextView 加入圓角和邊框

1. 建立資料文件，在 res -> drawable -> New Drawable Resource File -> 為 File 命名
2. 在 File 中替換如下

```
<?xml version="1.0" encoding="utf-8" ?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="#EBEDE38B"/>
    <stroke android:width="3dp" android:color="#4fa5d5"/>
    <corners android:radius="12dp" />
</shape>
```

```
// shape 控制圖形形狀
// solid 控制形狀填充色
// stroke 控制邊框粗度和顏色
// corners 控制圓角程度
```

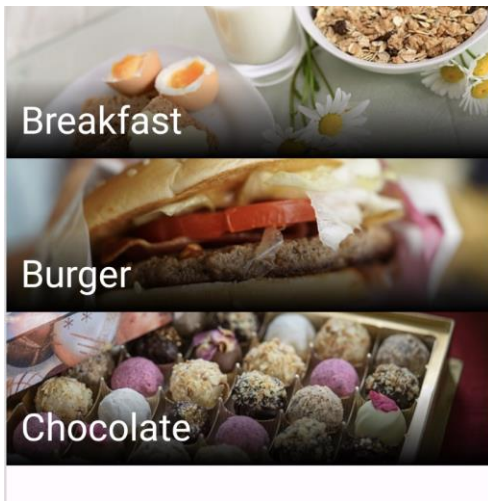
3. 為 TextView 加入背景 background 屬性

```
<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:background="@drawable/textview_default_bg"
    android:padding="12dp"
    android:text="TextView"
    android:textSize="32sp" />
```

用程式方法改變 textView 的背景和文字色

```
textView.setTextColor(Color.parseColor("#ff0000"))
textView.setBackgroundResource(R.drawable.drawables_name)
```

建立漸變背景



```
<shape xmlns:android="http://schemas.android.com/apk/res/android">  
  <gradient android:angle="90"  
    android:startColor="#2A2A2B"  
    android:endColor="#4D434343" />  
</shape>
```

android:startColor="#2A2A2B" //開始漸變顏色
android:endColor="#4D434343" //結束漸變顏色
android:angle="90" //漸變角度